

Problem:

- I'm trying to solve the problem of not knowing the temperature and not knowing environment trends in a room. People living in a place without a smart temperature and humidity monitor who want to track the trends in a room would use this device.

User Research

- I would need the temperature, humidity, and pressure.
- To get this I would need an OLED sensor.

Product Requirements

- Temperature/humidity/pressure
- OLED display
- SD-card data logging

Technical Implementation

- ESP32
- BME280
- OLED
- microSD
- microSD module
- PCB designed in KiCad

Product Decisions

- I chose the ESP32 because it has enough GPIO and processing capability to support multiple sensors and gave the project the option to connect to Wi-Fi eventually. I wanted the first version to focus on just sensing the environment while later version would log these readings.
- The BME280 combines three useful measurements in one sensor. This allowed me to collect pressure, temperature, and humidity with less hardware.
- I initially planned to use local storage on a SD card so that the environment monitor could log the data without internet connection. While developing this I had trouble getting the SD module to work so I prioritized the hardware of the sensor and screen over trying to make the Sd module to work.
- What I implemented vs. saved for later:

Priority	Feature	Reason
V1	BME280 sensing	Core purpose
V1	ESP32	Main controller
V1	OLED display	Displays results
V1	Custom PCB	Improves hardware from breadboard
V1- not working currently	SD data logging	Enables data logging
V2	Wi-Fi	Remote monitoring
V2	Firebase	Cloud storage
V3	Web dashboard	Visualization of the data
V3	alerts	Automated notifications